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Constraint handling: Maintaining feasibility / repairing infeasibility

Lecture 31: Presented by Jacomine Grobler

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In this final chapter, we yet again consider non-linear optimisation problems in which the objective is to

$$\text{maximise} \quad z = f(\boldsymbol{x})$$

subject to a set of constraints of the form

$$\begin{array}{rcll} g_j(\boldsymbol{x}) & \leq & a_j, & j = 1, \ldots, p, \\ h_k(\boldsymbol{x}) & = & b_k, & k = 1, \ldots, q, \\ \boldsymbol{x} & \in & \mathbb{R}^n, & \end{array}$$

where p and q are non-negative integers, where n is a natural number, where the functions f, $g_1, \ldots, g_p$ and $h_1, \ldots, h_q$ are not necessarily differentiable, and where $a_1, \ldots, a_p$ and $b_1, \ldots, b_q$ are real constants.

There are two main classes of constraint handling techniques:

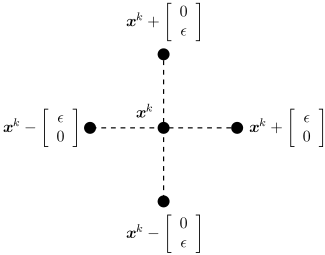
- Techniques that restrict the search process to the feasible region (maintain feasibility or repair infeasibility)
- Techniques that allow the search to venture into the infeasible region (penalty function)

Maintaining feasibility / repairing infeasibility

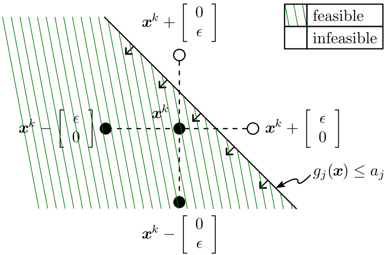
Four techniques for maintaining feasibility / repairing infeasibility:

- Ensure that perturbation operators cannot yield infeasible solutions (Example 7.1)
- Repeatedly reapply the perturbation operator until a feasible solution is obtained (Example 7.2)
- Repair infeasibilities with the closest point in the feasible search space (Example 7.3)
- Repair infeasibilities by replacing infeasible points with closest point on the hypersurface defined by the equality constraints (Example 7.4)

Maintaining feasibility



(a)



(b)

Example 7.1 (pp. 204–207)

Let us consider again the problem of determining the dimensions of a customised fish tank with a square bottom and an open top of largest possible enclosed volume, subject to the constraint of using at most four square metres of glass. Recall from Example 5.2 that the non-linear optimisation problem

$$\text{maximise } f_{7.1}\left(x_1, x_2\right) = x_1^2 x_2$$

subject to the constraints

$$\begin{aligned} x_1^2 + 4x_1x_2 &\leq 4 \\ x_1, x_2 &\geq 0 \end{aligned}$$

governs this decision, where x_1 denotes the floor length of the tank and x_2 denotes its height.

Example 7.2 (pp. 208–210)

Let us consider again the problem of determining the dimensions of a customised fish tank with a square bottom and an open top of largest possible enclosed volume, subject to the constraint of using at most four square metres of glass. Recall from Example 5.2 that the non-linear optimisation problem

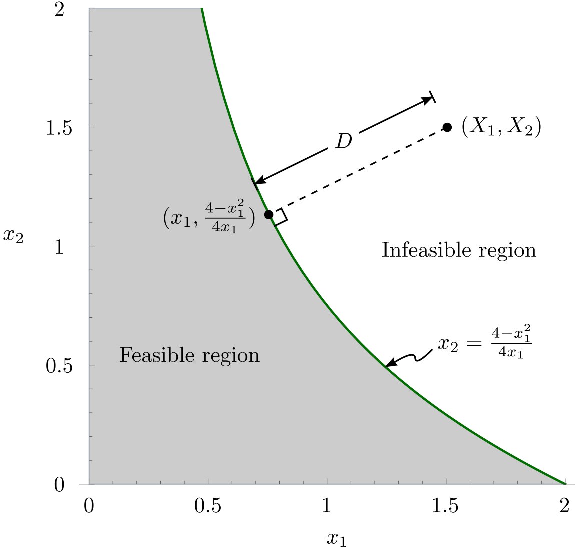
$$\text{maximise } f_{7.1}\left(x_1, x_2\right) = x_1^2 x_2$$

subject to the constraints

$$\begin{aligned} x_1^2 + 4x_1x_2 &\leq 4 \\ x_1, x_2 &\geq 0 \end{aligned}$$

governs this decision, where x_1 denotes the floor length of the tank and x_2 denotes its height.

Repairing infeasibility



Example 7.3 (pp. 210–217)

Let us consider again the problem of determining the dimensions of a customised fish tank with a square bottom and an open top of largest possible enclosed volume, subject to the constraint of using at most four square metres of glass. Recall from Example 5.2 that the non-linear optimisation problem

$$\text{maximise } f_{7.1}\left(x_1, x_2\right) = x_1^2 x_2$$

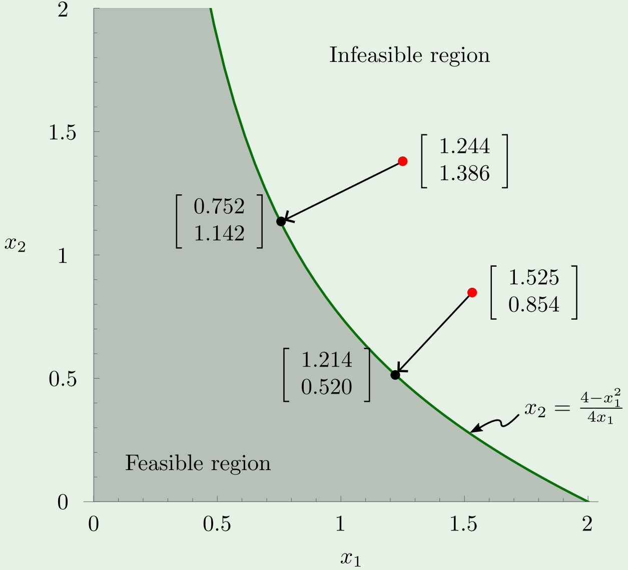
subject to the constraints

$$\begin{aligned} x_1^2 + 4x_1x_2 &\leq 4 \\ x_1, x_2 &\geq 0 \end{aligned}$$

governs this decision, where x_1 denotes the floor length of the tank and x_2 denotes its height.

Example 7.3 (pp. 210–217)

During iteration $k = 1$, the new position vectors $\boldsymbol{x}^1(1) = [0.251 \ \ 0.034]^T$, $\boldsymbol{x}^2(1) = [1.244 \ \ 1.386]^T$, $\boldsymbol{x}^3(1) = [0.051 \ \ 1.577]^T$, $\boldsymbol{x}^4(1) = [0.806 \ \ 0.272]^T$ and $\boldsymbol{x}^5(1) = [1.525 \ \ 0.854]^T$ are computed in Step 9 of Algorithm 4.7. Of these position vectors, the two typeset in red are, unfortunately, infeasible. Both these infeasible position vectors violate the constraint in (7.13).

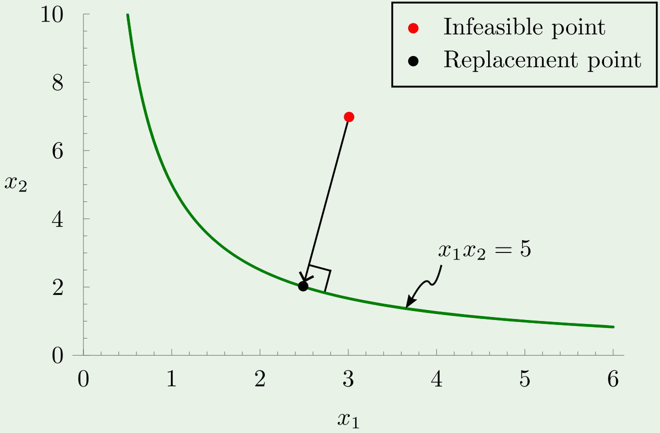

2
1.5
1
0.5
0
0
0.5
1
1.5
2
x_1
x_2
Infeasible region
Feasible region
$\begin{bmatrix} 1.244 \\ 1.386 \end{bmatrix}$
$\begin{bmatrix} 0.752 \\ 1.142 \end{bmatrix}$
$\begin{bmatrix} 1.525 \\ 0.854 \end{bmatrix}$
$\begin{bmatrix} 1.214 \\ 0.520 \end{bmatrix}$
$x_2 = \frac{4-x_1^2}{4x_1}$

Example 7.4 Consider the non-linear optimisation problem of

$$\text{maximising } f_{7.3}\left(x_1, x_2\right) = \exp\left[-\left(x_1 - 3\right)^2 - \left(x_2 - 2\right)^2\right]$$

subject to the constraints

$$\begin{aligned} x_1 x_2 &= 5 \\ x_1, x_2 &\geq 0. \end{aligned}$$


10
8
6
4
2
0
x_2
0
1
2
3
4
5
6
x_1
Infeasible point
Replacement point
$x_1x_2 = 5$

Example 7.4 (pp. 217–224)